

UML Diagrams Practice Problems

Question Patterns

Study the given user story and draw a UML diagram that

- Identifies the classes in the story with attributes, methods, and visibility,
- Shows generalization, association, aggregation, and composition,
- Includes multiplicity at both ends of every relationship,
- Marks interfaces, abstract and concrete classes where appropriate.

Tips:

It is very common for these problems to have more than one correct solution, and some solutions are often more correct than others. The trick is to look at the marks and estimate the effort required to answer the question. Always include the items listed above in your diagrams.

All parts of the diagram will likely remain connected. If you have multiple isolated parts, reread the story and find the missing link.

For questions or confusion, please refer to the respective course teacher for explanations and possible alternative representations.

Practice Problem #1

Community Fitness Center

FitLife operates community fitness centers. Each fitness center contains one or more rooms. A room is identified only within its fitness center; if the center is permanently removed from the system, its rooms are also removed.

A fitness center also keeps exercise equipment such as treadmills, exercise bikes, and weight machines. Equipment may be transferred from one fitness center to another, so it can continue to exist independently of a particular center.

The system stores information about members and trainers. Both members and trainers are people and share information such as name, phone number, and date of birth. A person in this system must be registered as either a member or a trainer.

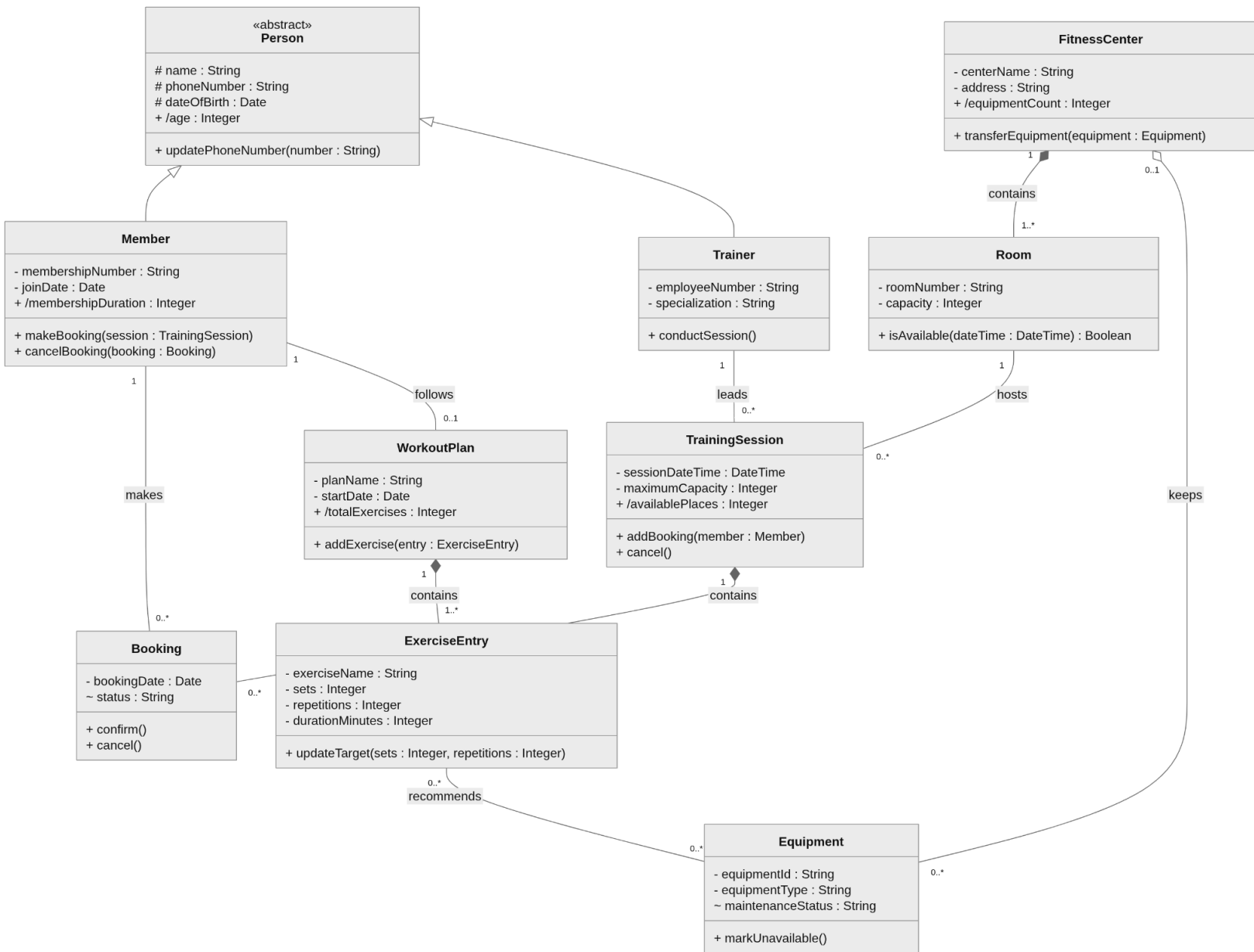
Trainers conduct training sessions. Every training session is led by exactly one trainer and takes place in exactly one room. A trainer may lead many sessions, while a room may host many sessions at different times.

Members reserve places in training sessions. Each reservation is represented by a booking. A booking belongs to exactly one member and exactly one training session. A member may make many bookings, and a session may have many bookings as well. If a training session is canceled and removed, all its bookings are also removed.

A member may have one active workout plan, though newly registered members may not yet have one. Each workout plan belongs to exactly one member.

A workout plan contains one or more exercise entries. An exercise entry records details such as the exercise name, number of sets, repetitions, and duration. An exercise entry cannot exist without its workout plan. An entry may optionally recommend one or more pieces of equipment, and the same equipment may be recommended by many entries.

Practice Problem #1 Solution



Practice Problem #2

Dhaka Public Library Network

The Dhaka Public Library Network manages several library branches located in areas such as Dhanmondi, Uttara, Mirpur, and Motijheel. Every branch belongs to the network. If a branch is permanently removed from the system, its branch-specific information is also removed.

Each branch maintains a collection of library items. Items can be transferred between branches; therefore, a library item can continue to exist even if its current branch closes. Every item has a unique item code, title, acquisition date, and availability status. It provides three types of items:

- A book has an ISBN, an author, and a page count. Patrons may reserve a book when it is unavailable.
- A DVD has a director and running time.
- A journal has an issue number, publication date, and one or more articles. An article belongs to a particular journal issue and cannot exist in the library catalog without that journal.

People registered with the library become patrons. Each patron has a library card number, name, phone number, address, and registration date. The patron's membership duration is calculated from the registration date.

A patron may borrow multiple library items. Every borrowing transaction is recorded as a loan. Each loan is associated with exactly one patron and exactly one library item. It records the issue date, due date, and, when applicable, the return date. The overdue status and fine are calculated rather than stored.

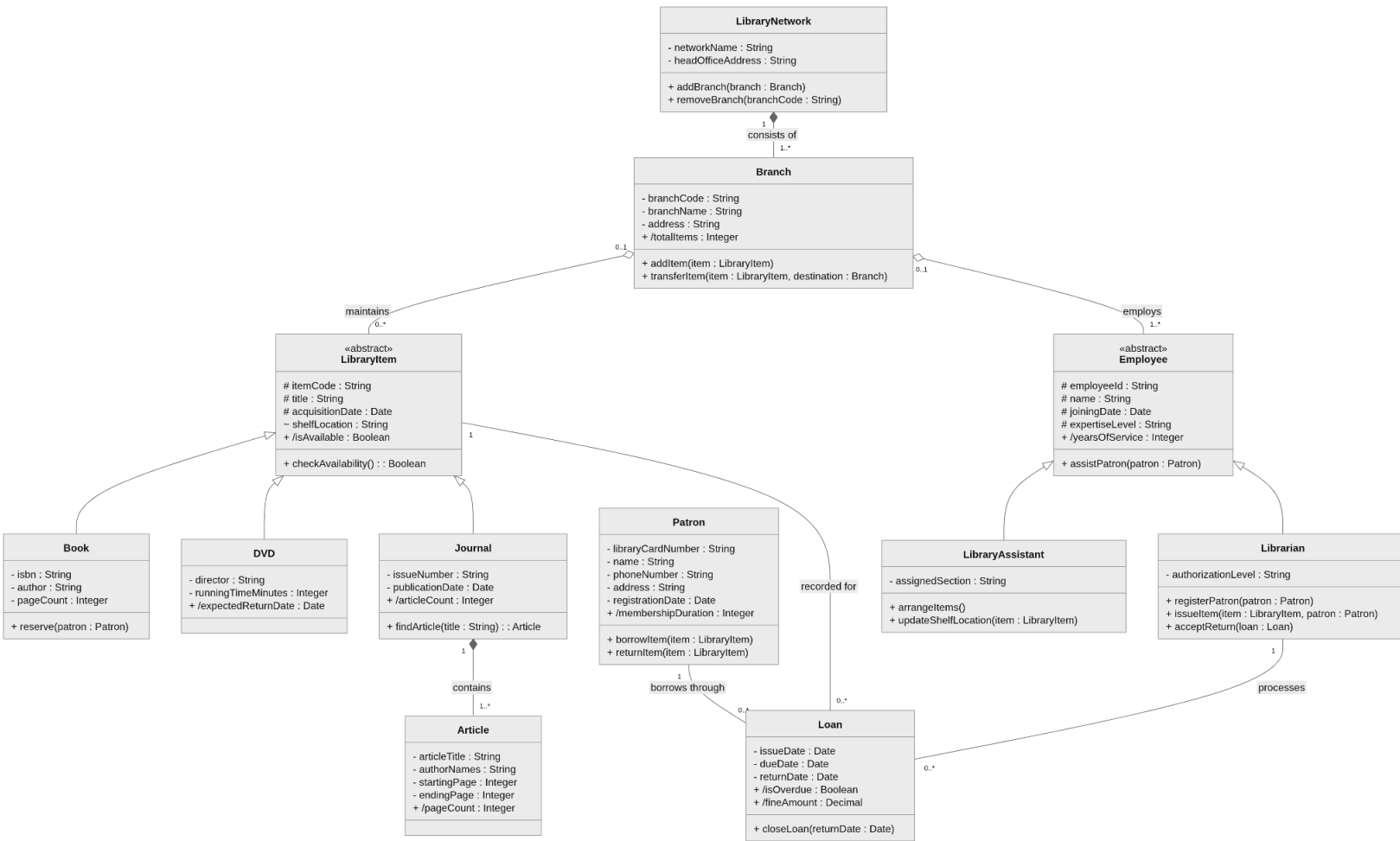
An item can have many loan records over its lifetime, but at most one active loan at a time. A newly registered patron may not have borrowed anything yet.

Each branch has employees. Employees may be transferred between branches and therefore exist independently of a particular branch. Every employee has an employee ID, name, joining date, and expertise level. There are two types of employees:

- **Librarians:** register patrons, issue items, accept returned items, and manage reservations.
- **Library assistants:** arrange items, update shelf locations, and help patrons find resources.

A branch must have at least one librarian and may have any number of library assistants.

Practice Problem #2 Solution



Practice Problem #3

GoalBangla World Cup Analytics

GoalBangla is a Bangladeshi sports analytics platform that tracks the ongoing Football World Cup and generates match predictions for local viewers. Match times are displayed in Bangladesh Standard Time.

An administrator registers the participating national teams and assigns each team to a tournament group. A group contains several teams, but teams exist independently and may be reassigned during tournament setup.

Each team registers multiple players. A player remains in the platform even when removed from a team's World Cup squad.

Every match is played between exactly two teams. The system records its stage, kickoff time, match status, and final scores. The winner, total goals, and goal difference are calculated from the scores.

The platform stores two types of match statistics:

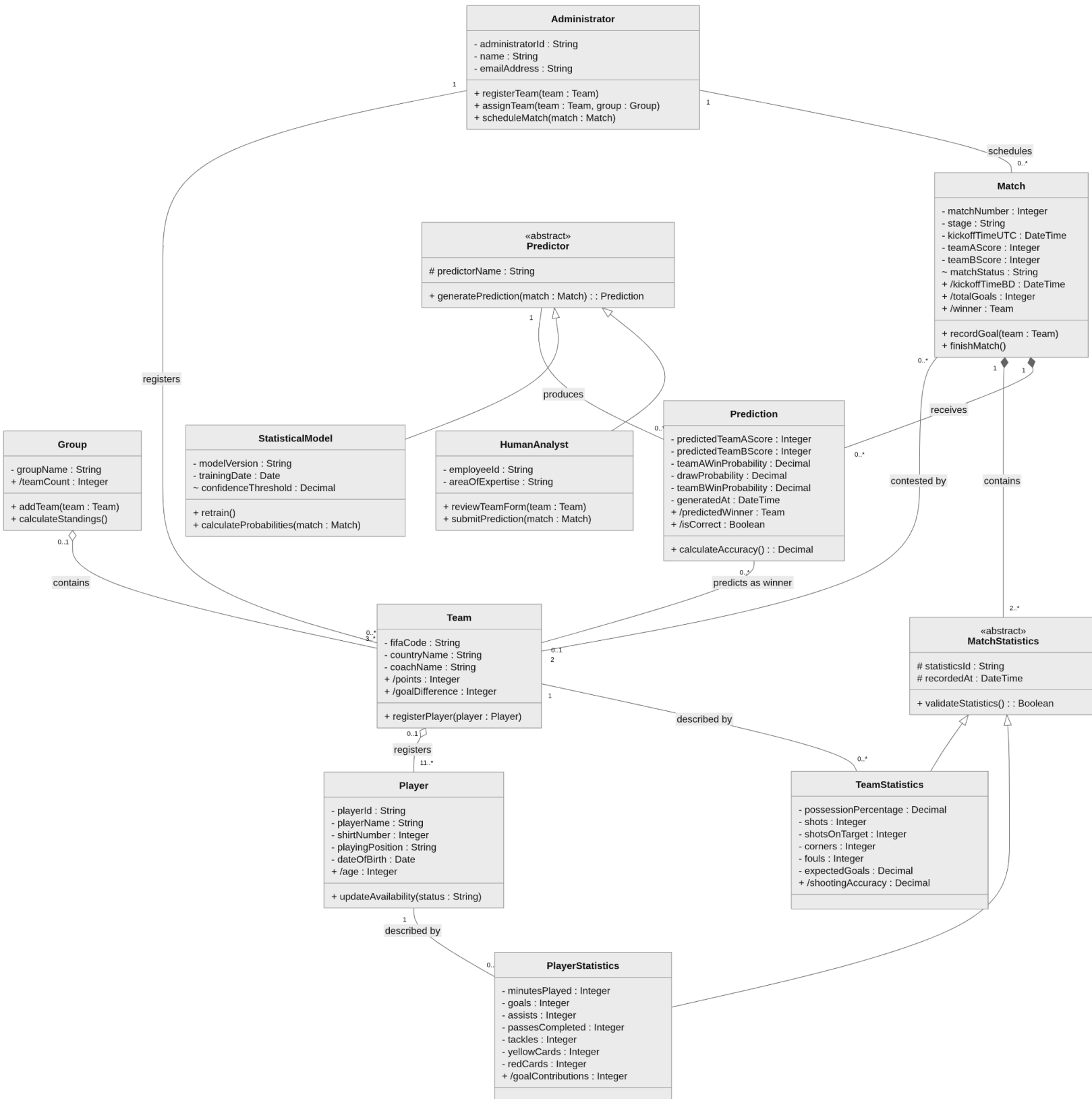
- Team statistics include possession, shots, corners, fouls, and expected goals.
- Player statistics include minutes played, goals, assists, passes, tackles, and cards.

Statistics belong to a particular match and are removed if that match is deleted.

Before a match, GoalBangla creates predictions that include the predicted score and the probabilities of a team win, a draw, or an opponent win. After the match, the system calculates whether each prediction was correct.

A prediction may be generated by a statistical model or submitted by a human analyst. Every prediction belongs to exactly one match and is produced by exactly one predictor. A match may have predictions from several models and analysts.

Practice Problem #3 Solution



Practice Problem #4

Texas Hold'em Game Simulator

A university programming club in Bangladesh is developing an online Texas Hold'em simulator for friendly competitions. The game uses virtual chips only and does not involve real money.

A poker game has between two and eight players. Players exist independently and may join different games. Every player has a name and a stack of virtual chips. There are two types of players:

- A human player has a username and connection status.
- A computer player has a difficulty level: easy, medium, or hard.

Every game has one dealer responsible for starting rounds and managing cards. A game consists of multiple game rounds. If a game is deleted, its rounds are also deleted.

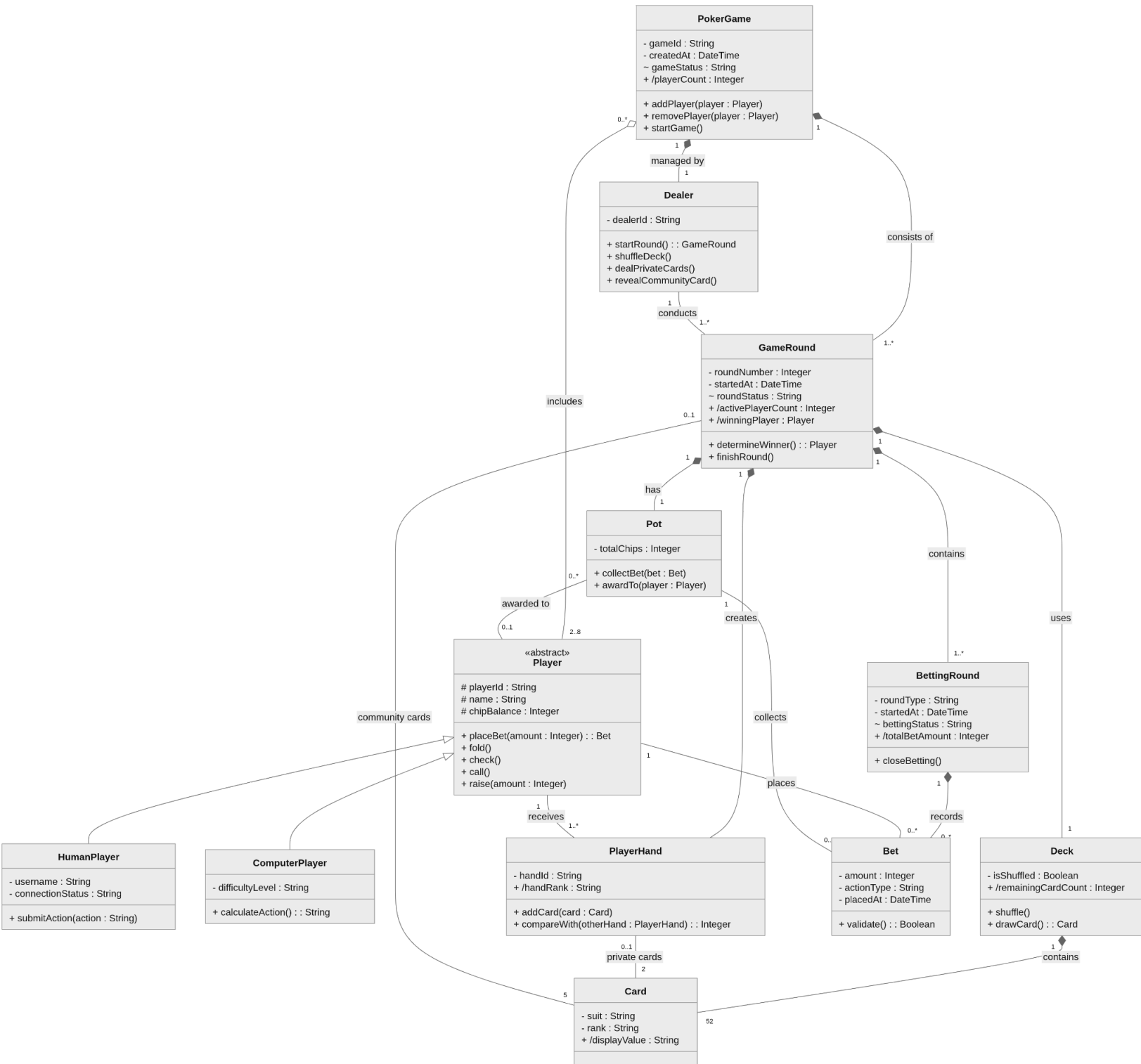
At the beginning of each round, the dealer creates and shuffles a standard deck of 52 cards. Each card has a suit and rank and belongs to that deck. The dealer gives each participating player a player hand containing exactly two private cards.

The round also has five shared community cards. The dealer initially reveals three community cards, known as the flop, followed by the turn and river cards.

Several betting rounds take place. Each betting round records the bets made by players. A player may fold, check, call, or raise. Every bet belongs to exactly one betting round and is placed by exactly one player.

All chips bet during a game round are collected in a pot. At the end, if more than one player remains, their hands are compared. The player with the best hand wins the pot.

Practice Problem #4 Solution



Practice Problem #5

Bangladesh Stock Portfolio App

A Bangladeshi fintech company is developing ShareTrack, a portfolio application for investors who trade securities listed on the Dhaka Stock Exchange or Chittagong Stock Exchange.

An administrator registers stock exchanges, listed companies, and their stocks. Each stock is issued by exactly one company and traded on exactly one exchange. A company may issue more than one type of stock.

The application stores daily price records for every stock, including opening, closing, highest, and lowest prices in Bangladeshi taka. A price record belongs to one stock and has no meaning without it.

An investor may create multiple portfolios for different investment goals. A portfolio belongs to exactly one investor and contains holdings. A holding records the quantity and average purchase price of one stock. Its current market value and profit or loss are calculated from the latest stock price.

Investors may place buy or sell orders through a portfolio. Every order is for exactly one stock and records its quantity, requested price, date, and status. Buy and sell orders follow the same basic order structure but perform different validations.

A portfolio may contain many orders and holdings. However, it cannot contain more than one holding for the same stock.

Practice Problem #5 Solution

